

Lea Brody-Heine

lea_brody-heine@alumni.brown.edu | Software Engineer | **BA & MSc**
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Software engineer with 2+ years of experience delivering production software, secure RAG systems, test automation, and developer tooling. My work spans on-call operations, code reviews, debugging, and production reliability, as well as cross-functional development and end-to-end project ownership.

Education

University of St Andrews *MSc Computer Science*

GPA: First Class Honors with Distinction | Dean's List

St Andrews, Scotland

Graduated: 09/24

Brown University *BA in International Affairs*

GPA: 4.0

Providence, RI

Graduated: 05/23

Work Experience

Google

12/24 - 07/26

Software Engineer | Ads

Sunnyvale, CA

- Delivered an end-to-end manual-release testing automation platform from 0-to-1 in 9 months, aligning 3+ cross-functional stakeholder teams on a shared rollout timeline and authoring 4 technical design documents and proposals; eliminated a 24-hour manual bottleneck, achieved 100% test coverage, and saved \$60K annually.
- Delivered a performance-metrics data pipeline from design through launch by building a custom device-metrics crawler and parser, automating jank, memory, and latency tracking and cutting manual triage time by 30%.
- Built an agentic test-analysis and stress-testing pipeline across 5 ad formats that autonomously analyzed logs and surfaced SDK breaking points.
- Cut the failing-test backlog by 50% (130 to 65) and processed 500+ bug reports through daily Python-based triage automation.
- Supported production reliability through on-call infrastructure rotations, incident debugging, code reviews, and recurring production-metrics reporting.

Paid Independent Contracts

Remote

- **GSI Water Solutions, Inc. (09/26 - Present):** Developing a secure RAG prototype for private-document search with Python, FastAPI, React/TypeScript, PostgreSQL/pgvector, and locally computed BGE embeddings, using Oregon water law as the initial validation domain.
 - Implementing SQL-enforced group permissions, full-text/vector hybrid retrieval with reciprocal-rank fusion, cited answers across deterministic, Ollama, and Anthropic backends, and an evaluation harness to track retrieval quality as the system evolves.
- **AIM Executive Coaching (09/24 - 10/24):** Built and launched a responsive client website with HTML, CSS, JavaScript, and HubSpot, increasing discoverability by 21% through SEO, performance optimization, and cross-device development.
 - Translated requirements and wireframes into a multi-page product with responsive navigation, service pages, interactive coaching and speaking modals, testimonial sliders, and newsletter and contact flows; iterated directly with the client on usability.

Relevant Projects

- **AI Production Evaluation Platform:** Built a cloud-native RAG evaluation platform spanning four FastAPI services and a React/TypeScript client, with PostgreSQL/pgvector, Redis caching, Docker Compose, and Kubernetes deployment paths.
 - Implemented bounded agentic retrieval, cited answers, interchangeable deterministic, vLLM, and Triton backends, quality regression gates, load-test scenarios, and OpenTelemetry/Prometheus/Grafana observability; maintained 477 Python and 10 web tests.
- **CI/CD Failure Triage Agent:** Built a Python agent that polls GitHub Actions, classifies failures as flakes, regressions, infrastructure issues, or new bugs with Claude, and generates root-cause hypotheses grounded in cited log evidence.
 - Automated confidence-gated, deduplicated GitHub issue filing with retry handling, an idempotent SQLite audit trail, a classifier evaluation harness, a self-contained HTML observability report, and comprehensive pytest coverage.
- **Desktop Calendar Reminder App:** Built and use a native macOS menu-bar app daily for personal calendar reminders; implemented it in Swift/SwiftUI with EventKit and AppKit to display animated, non-activating alerts for selected Apple and synchronized Google calendars.
 - Implemented configurable reminder intervals, duplicate-safe scheduling, grouped simultaneous events, local settings, launch-at-login, click-through to Calendar, wake and calendar-change recovery, and XCTest coverage.

Technical Skills

Programming: Python, Java, Swift, TypeScript, JavaScript, Kotlin, SQL, HTML, CSS

Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, System Design, RESTful APIs, Distributed Systems, Multithreading, Concurrency, Debugging, Code Reviews, Developer Tooling, API Design, Backend Development, Full-Stack Development, Android, iOS, Mobile Ads

AI & Data: Retrieval-Augmented Generation (RAG), LLM Integration, Agentic Workflows, Embeddings, Vector Search, Hybrid Search, PostgreSQL/pgvector, Data Pipelines, Data Processing

Testing & Reliability: Test Automation, Unit Testing, Integration Testing, End-to-End Testing, Performance Testing, Load Testing, Regression Testing, CI Failure Triage, Release Engineering, Production Reliability, On-Call Support, Incident Debugging, Observability

Frameworks & Databases: FastAPI, React, Node.js, Express, SwiftUI, PostgreSQL, Redis, SQLite, MongoDB, MySQL

Tools & Platforms: Git, GitHub Actions, CI/CD, Docker, Kubernetes, Linux, Bazel, Cloud APIs, Containerization, Version Control